

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester VI Examination, 2022-23

Paper: GLB-604 — Economic Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 058443

*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory. The figures in the right-hand margin indicate marks.*

SECTION — A

Question 1. What is cavity filling? Explain the characteristic features of cavity filling with suitable sketches. [14]

Question 2. Write about the origin, modes of occurrence, and Indian distribution of any two minerals used in the chemical industry. [14]

Question 3. Write short notes on **any two** of the following: [2 × 7 = 14]

- (a) Classification of gemstones
- (b) Raw materials for phosphate in the fertilizer industry
- (c) Residual liquid injection

Question 4. What is a placer deposit? What are placer minerals? Discuss the formation of placer deposits. [14]

SECTION — B

Question 5. Define ore body in a mineral deposit. Describe various morphologies of ore bodies in mineral deposits with suitable diagrams. [14]

Question 6. Describe the mineralogy, modes of occurrence, economic uses, and distribution of iron ore deposits in India. [14]

Question 7. What is petroleum? Briefly describe the geology, modes of occurrence, and distribution of petroleum deposits in India. [14]

SECTION — C

Question 8. Answer the following:

- (a) What is paragenesis? [2]
- (b) Define tenor of ore with an example. [2]

- (c) Define the physical constituents of coal. [2]
- (d) Write the name of an important ore mineral of tin. [1]
- (e) What is gossan? [2]
- (f) Write the names of any two organic materials used as gems. [2]
- (g) What is refractory? [2]
- (h) Why is wollastonite used in the ceramic industry? [1]